

Signal Processing and Fourier Transforms

Fourier Transform decomposes a signal into frequency components. FFT is an efficient $O(n \log n)$ algorithm for discrete signals. Nyquist theorem: sampling rate must exceed twice the highest frequency to avoid aliasing. Low-pass filters remove high frequencies; high-pass filters remove low ones. Convolution in time domain equals multiplication in frequency domain. Used in audio, image, and communication systems.